

Exercise3

Duxing Li · 29 de marzo de 2026



Exercise 3

Three persons work independently to decipher a message. The probabilities that they will decipher the message are $\frac{1}{5}$, $\frac{1}{4}$ and $\frac{1}{3}$. Calculate the probability that the message will be deciphered? Verify your calculation by simulation. (Ans. 0.60)

#Solution here **Solution:**

Since the three persons work independently, So they are independent.

1 minus the situation of all fails

$$\begin{aligned} P &= 1 - P(\text{all fail}) \\ &= 1 - \left(1 - \frac{1}{5}\right)\left(1 - \frac{1}{4}\right)\left(1 - \frac{1}{3}\right) \\ &= 1 - \left(\frac{4}{5} \cdot \frac{3}{4} \cdot \frac{2}{3}\right) = 1 - \frac{2}{5} = \frac{3}{5} = 0.6 \end{aligned}$$

In [3]:

```
from numpy.random import default_rng

rng = default_rng()

n_trials = 100000
count = 0

for i in range(n_trials):
    p1 = rng.choice(['success', 'fail'], p=[1/5, 4/5])
    p2 = rng.choice(['success', 'fail'], p=[1/4, 3/4])
    p3 = rng.choice(['success', 'fail'], p=[1/3, 2/3])

    if p1 == 'success' or p2 == 'success' or p3 == 'success':
        count += 1

print("Simulated probability =", count / n_trials)
print("Exact probability =", 0.6)
```

Out [3]:

```
Simulated probability = 0.59793
Exact probability = 0.6
```